UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2009 question paper for the guidance of teachers

5090 BIOLOGY

5090/02 Paper 2 (Theory), maximum raw mark 80

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UNIVERSITY of CAMBRIDGE
International Examinations

Section A

1 (a) asexual/vegetative ; [1]

(b) no variation or same genes as parent / no evolution / overcrowding / susceptible to disease / no resistance to environmental change ; [1]

(c) *one mark per line, mark the first given* :

no wastage / offspring well established before separating (**R** chancy unqualified) ;

quick / only one parent needed ;

offspring of known characteristics / genetically identical AW ;

more profit AW ; [max 2]

(d) **(D)** photosynthesis ;

manufacture CHO or named CHO / amino acids (**R** food) ; [2]

(E) absorb water ;
ions / salts / minerals (**R** nutrients) ;

anchorage ; [max 2]

(F) carries / transports AW + *water / *salts ;

before roots develop AW ;

carries / transports AW + *sugars (**R** glucose) / *amino acids ;

correct functional ref. either xylem or phloem ; [max 3]
(* or allow 'nutrients' once only if neither of these marks is otherwise scored)

[Total: 11]

2 (a) *For each letter, if more than one cell labelled, all labels must be correct*

(i) label line (or otherwise indicated) **J** to / in / on cellulose cell wall
(**R** label lines that terminate on inner line of cellulose cell wall) ;

(ii) label line (or otherwise) **K** to cytoplasm ; [2]

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(b) solution G (**A** under explanation if solution line blank) ; [1]

water enters ;

*osmosis / diffusion ;

*cell sap more concentrated / diffusion or osmotic or water potential gradient ;

increase in volume AW / pressure / turgor ; [max 4]
(* available with wrong or no solution identified)

(c) (i) space between membrane and cell wall shaded on right hand diagram ; [1]
(ignore shaded cell wall)

(ii) (*These marks are available with no or wrong shading if they are in reasonable context*)

cell wall permeable ;

solution passes in ;

correct ref. to partial permeability of cell membrane ; [3]

[Total: 11]

3 (a) (L) right + atrium / auricle ;

(M) pulmonary vein ; [2]

(b) (chemicals) *glucose ;

*oxygen ; [2]

(explanation max 2 – each mark available once only either under glucose or oxygen)

for energy (**R** production) ;

during respiration ;

heart muscle / cells / tissue ; (max 2) [4]

(***A** adrenalin; to stimulate heart muscle; insulin; for glucose absorption by heart cells; Heparin; to stop blood clotting)

(c) valve (if named, must be aortic / semi-lunar) ;

closes (available with wrongly-named valve) ;

sending blood through coronary blood vessel / artery(ies) (**R** vein) ; [3]

[Total: 9]

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4 **(a)** fruit / pericarp / ovary / testa ; [1]

(b) wind / moving air ; (1)

*(**R** any refs. to water dispersal)*
ref parachute / helicopter wing AW ;

large surface area / light ;

buoyancy / holds seed in air or floats ; (max 2) [3]

(c) lack of water AW ;

lack of O_2 (for whatever reason) ;

seed not viable / dormancy not complete AW / eaten ; [max 2]

(d) overcrowding / lack of light ;

unsuitable temperature ;

eaten ;

disease ;

unsuitable substrate / lack of ions or minerals / wrong pH ;

(**A,** here, ref. insufficient nutrients in soil = unsuitable substrate)

drought AW ; [max 3]

[Total: 9]

5 **(a)** photosynthesis ; [1]

(b) **(i)** 20.00 (hrs) / 8 p.m ; [1]

(ii) photosynthesis slows down ;

photosynthesis stops ;

respiration occurring (**R** respiration starts) ;

carbon dioxide released from plant ; [max 3]

(c) (increased) temperature ; [1]

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(d) curve starting below 0 ;

crossing 20.00 hr line at 0 ;

finishing above 0 at 24.00 hrs ;

axis labelled correctly for line drawn (Ignore units) must be on Fig. 5.1 ;

If line is straight, only crossing and axes marks available.

***A** curve 'upside down' with axis labels reversed.*

Last point available only on Fig. 5.1, award otherwise what is possible on a graph drawn elsewhere. [4]

[Total: 10]

[Section A Total: 50]

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Section B

6 **(a)** chemical (**R** named hormone) ;

produced by a gland (**R** specific named endocrine gland) ;

carried + by the blood ;

affects a target organ (**R** specific target organ) ;

destroyed by liver ; [max 4]

(b) stimulus or example ;

receptor / or described (**R** named sense organs) ;

impulse / electrical pulse AW ;

sensory neurone or sensory nerve cell (**R** nerve) ;

correct ref. to CNS for action described ;

relay / intermediate neurone or nerve cell (**A** 'interneurone') ;

motor or effector neurone or nerve cell ;

effector or described (muscle or gland) ;

correct response for stimulus given / named reflex ; [max 6]

[Total: 10]

7 **(a)** in nucleus / plasmids / chloroplasts / mitochondria ;

found in chromosomes / genes / contains genetic information ;
(**R** contains genes)

can be copied / inherited / ref responsible for characters (characteristics) ;

controls production of a protein ; [max 3]

(b) mutation (with ref. either sickle cell anemia or Down's) ;

(sickle cell) genetic or ref. gene / allele / DNA / change in Hb molecule ;

(Down's syndrome) one extra chromosome / 47 instead of 46 ; [3]

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(c) correct plausible genotypes of parents (**A** A,B,O,I^A,I^B,I^o only) ;
(if they are linked by lines to the offspring, since they are then gametes and available below)

phenotypes / blood groups of both parents (**A** 'B' as Group B etc.) ;

gametes correctly shown – either separated / in circles etc. / punnet square / linked correctly by lines to offspring ;

offspring correctly identified + with different stated blood group / phenotype ;

any correct descriptive term (gametes / parents / genotype / children) ; [max 4]
(**A** mother + father)

(*Allow all marks on a written description)*

[Total: 10]

8E (a) soil erosion / washed / blown ;

loss of humus in soil ;

desertification / ref. less rainfall / less transpiration ;

ref. leaching / flooding / loss of soil fertility ;

loss of species / habitats / qualified effect on food chains ;

loss of livelihood / resources / agricultural effects ;

global warming AW / CO_2 increase / climate change AW ; [max 5]

(b) insecticides + specific undesirable effect ;

fertilisers / sewage / domestic or nitrogenous waste + specific undesirable effect ;

heavy metal / inorganic, chemical or radioactive waste + specific undesirable effect ;

gases from factories, car exhaust (Pb) or fossil fuel / SO_2 / NxOy / soot
+ specific undesirable effect ;

CFCs / CO_2 / methane / CO + specific effect ;

oil (spillage) + effect on wildlife ;

litter / rubbish / noise / light + effect ; [max 5]

[Total: 10]

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8O (a) light + producer / green plant AW ;

each organism uses energy ;

or loses energy ;

none returned to producer / sun ; **[max 3]**

(b) (i) food web is interlinked food chains / ref. many consumers [1]

(ii) producer + plants (or named) ;

Consumers ;

herbivores / primary or first order ;

carnivores / secondary or second order ;

ref. to relative numbers ;

ref. to relative biomasses ;

ref. to each trophic level controlling the numbers in neighbouring trophic levels ;

ref. to members all being in the same habitat / ecosystem ; [max 6]

[Total: 10]

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